Lab 5: Ohm's Law

March 7, 2026

Jaysen Tsao

Lab Partner: Alex Yue

PHYS 262 CRN 30949

Professor Szwerc

Jaysen Tsao
PHYS 262 CRN 30949

Pre-lab Questions

1. A resistor with a green-blue-orange-gold color code has resistance:

$$R = 56 \cdot 1\ \text{k}\Omega \pm 5\% = \mathbf{56\ k\Omega \pm 2.8\ k\Omega}.$$

2. If R_1, R_2, R_3 are the resistances of three resistors in parallel, their equivalent resistance is:

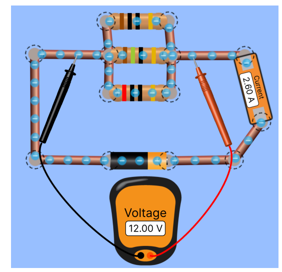
$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \quad \rightarrow R_{\text{eq}} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)^{-1}.$$

If they are instead connected in series, then:

$$R_{\text{eq}} = R_1 + R_2 + R_3.$$

In the PhET Lab, I used three resistors with resistances $R_1 = 10\ \Omega$, $R_2 = 15\ \Omega$, $R_3 = 20\ \Omega$.

- In parallel, the voltmeter measured a potential difference of 12.00 V across the resistors, and the ammeter measured a current of 2.60 A flowing through the circuit. Thus, the equivalent resistance of the parallel combination is:



$$R_{\text{eq}} = \frac{\Delta V}{i} = 4.61\ \Omega.$$

Using the equation from above, the calculated equivalent resistance of the parallel combination is:

$$R_{\text{eq}} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)^{-1} = \left(\frac{1}{10\ \Omega} + \frac{1}{15\ \Omega} + \frac{1}{20\ \Omega}\right)^{-1} = 4.62\ \Omega.$$

This is consistent with the measured value.

- In series, the voltmeter measured $\Delta V = 12.00$ V across the resistors, and the ammeter measured $i = 0.27$ A:

$$R_{\text{eq}} = \frac{\Delta V}{i} = 44.44\ \Omega.$$

Using the equation from earlier:

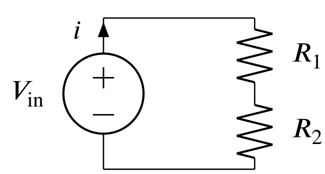
$$R_{\text{eq}} = R_1 + R_2 + R_3 = 10\ \Omega + 15\ \Omega + 20\ \Omega = 45\ \Omega.$$

This is consistent with the measured value.

3. Using Kirchhoff's Loop Rule, the current through the circuit is:

$$+V_{\text{in}} - iR_1 - iR_2 = 0$$

$$i\left(R_1 + R_2\right) = V_{\text{in}} \rightarrow i = \frac{V_{\text{in}}}{R_1 + R_2}.$$



The potential difference across R_1, $V_{\text{out}}\left(R_1\right)$, is:

$$V_{\text{out}}\left(R_1\right) = iR_1 = \frac{V_{\text{in}} R_1}{R_1 + R_2}.$$

Jaysen Tsao
PHYS 262 CRN 30949

Part 1: Resistor Color Code, Voltage, and the DMM

	Color Code Resistance	**Measured Resistance**	**Within Tolerance?**
R_1	$510\,\Omega \pm 5\%$	$517\,\Omega$	Yes
R_2	$75.0\,\mathrm{k}\Omega \pm 5\%$	$74.4\,\mathrm{k}\Omega$	Yes

Part 2: Ohm's Law Graph and Power Calculations

Color Code Resistance	1500 Ω
Measured Resistance	1479 Ω

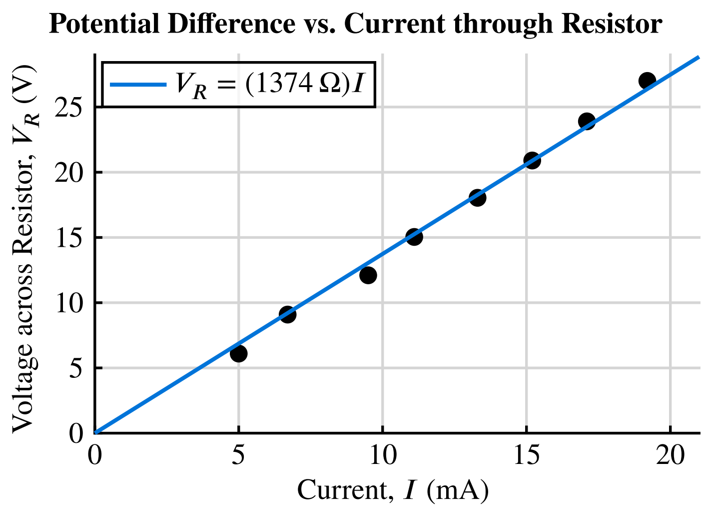
Safe Voltage Calculation

Let $P_{\max} = 0.5$ W. Since $P = V^2/R \to V = \sqrt{PR}$,

$$V_{\max} = \sqrt{P_{\max} R} = \sqrt{(0.5\text{ W})(1479\,\Omega)} = 27.4\text{ V}.$$

Data Collection & Analysis

Voltage V_R	Current I	Power $P = V_R I$
6.10 V	5.000 mA	30.50 mW
9.09 V	6.700 mA	60.90 mW
12.1 V	9.500 mA	115.0 mW
15.0 V	11.10 mA	166.9 mW
18.0 V	13.30 mA	239.9 mW
20.9 V	15.20 mA	317.7 mW
23.9 V	17.10 mA	408.7 mW
27.0 V	19.20 mA	518.4 mW



The slope of the best-fit line through the origin is 1374 Ω. Since $V_R = RI$ by Ohm's Law, the slope estimates the resistance R. Thus, $R \approx 1374\,\Omega$.

Resistance from Slope	1374 Ω
Measured Resistance	1479 Ω
Percent Difference	7.37%

Part 3: Resistors in Series, Parallel, and Series/Parallel

Resistor	Color Code Value	Measured Resistance
R_1	$220\,\Omega \pm 5\%$	$217\,\Omega$
R_2	$820\,\Omega \pm 5\%$	$821\,\Omega$
R_3	$1000\,\Omega \pm 5\%$	$979\,\Omega$

Series (R_1 and R_2)

$$R_{\text{series}} = R_1 + R_2 = 217\,\Omega + 821\,\Omega = 1038\,\Omega.$$

Calculated R_{series}	$1038\,\Omega$
Measured R_{series}	$1039\,\Omega$
Percent Difference	0.10%

Parallel (R_1 and R_2)

$$R_{\text{parallel}} = \left(\frac{1}{R_1} + \frac{1}{R_2}\right)^{-1} = \left(\frac{1}{217\,\Omega} + \frac{1}{821\,\Omega}\right)^{-1} = 171.634\,\Omega.$$

Calculated R_{parallel}	$171.634\,\Omega$
Measured R_{parallel}	$172\,\Omega$
Percent Difference	0.21%

Series/Parallel (R_1 and R_2 in series, R_3 in parallel)

$$R_{\text{eq}} = \left((R_1 + R_2)^{-1} + R_3^{-1}\right)^{-1} = \left((217\,\Omega + 821\,\Omega)^{-1} + \frac{1}{979\,\Omega}\right)^{-1} = 503.818\,\Omega.$$

Calculated R_{eq}	$503.818\,\Omega$
Measured R_{eq}	$504\,\Omega$
Percent Difference	0.04%

Part 4: Voltage Divider w/ Potentiometer

The potentiometer has a total resistance of $R_{tot} = 10\,\text{k}\Omega$, and $V_{in} = 10\,\text{V}$.

$$V_{out(R_1)} = V_{in} \cdot \frac{R_1}{R_1 + R_2}.$$

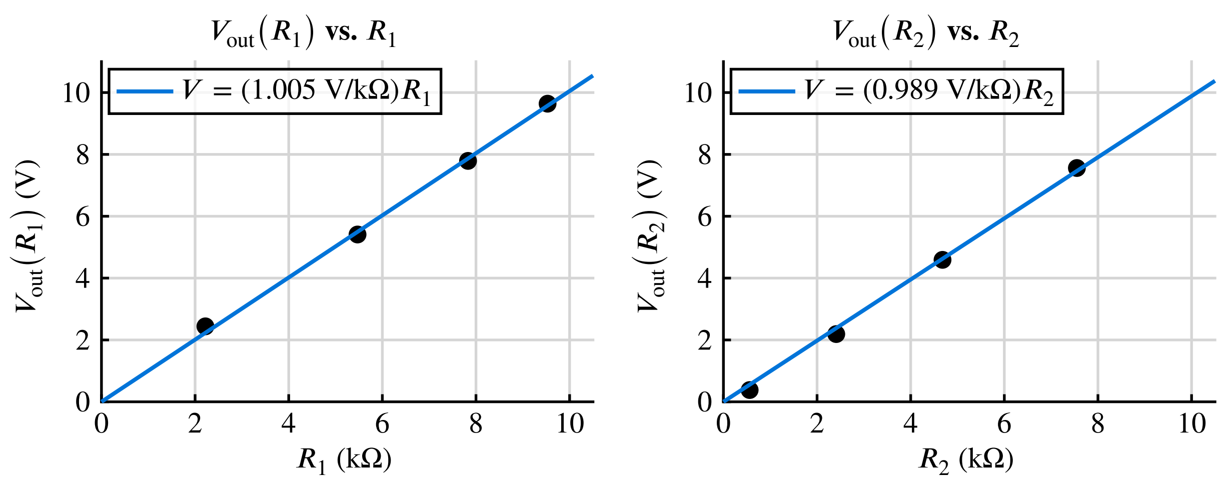
Demonstration: $R_1 + R_2 = 10\,\text{k}\Omega$

Position	R_1	R_2	$R_1 + R_2$
Left	$0\,\Omega$	$9750\,\Omega$	$9750\,\Omega$
Center	$3150\,\Omega$	$6960\,\Omega$	$10\,110\,\Omega$
Right	$9730\,\Omega$	$0\,\Omega$	$9730\,\Omega$

In all three positions, $R_1 + R_2 \approx 10\,\text{k}\Omega$.

Data Collection & Analysis

R_1	$V_{out}(R_1)$	R_2	$V_{out}(R_2)$
2.22 kΩ	2.44 V	7.55 kΩ	7.56 V
5.47 kΩ	5.41 V	4.68 kΩ	4.59 V
7.83 kΩ	7.79 V	2.41 kΩ	2.19 V
9.53 kΩ	9.64 V	0.56 kΩ	0.38 V



Using the voltage divider formula from the pre-lab:

$$V_{out}(R_1) = \frac{V_{in}}{R_{total}} \cdot R_1 = \frac{10\text{ V}}{10\text{ k}\Omega} \cdot R_1 = (1.000\text{ V/k}\Omega)R_1.$$

If V_{out} is treated as a function of R_1, the slope of the graph should be around 1 V/kΩ. Indeed, in the graph for R_1, the slope of the line of best fit is 1.005 V/kΩ which is very close to the expected result. Qualitatively, this means the voltage output through R_1 is proportional to the resistance R_1, which can vary as I turn the knob. The same logic can be applied to R_2: the slope of 0.989 V/kΩ is very close to 1 V/kΩ.

Jaysen Tsao
PHYS 262 CRN 30949

Conclusion Questions

1. The circuit is called a voltage divider. Why?

   The circuit splits, or "divides", the voltage among each of the two resistors. This allows two parts of the circuit to receive different parts of the total supplied V_{in}.

2. Does it represent a practical method of getting low voltages from very high voltages? Why or why not? (Hint: Think about power being lost or wasted by R_1 or R_2.)

   I dont think this method is practical for large voltage drops. Power is dissipated in R_1 and R_2 at a rate $P = V^2/R$, so for large input voltages, energy is wasted as heat at a significant rate (the rate increases by the square), which would cause a significantly lower V_{out}.

3. The DMM has an input impedance (resistance) of $10\,\text{M}\Omega$, so it does not draw an appreciable amount of current from this circuit. What would happen if the input resistance of the voltmeter were about $25\,\text{k}\Omega$? How would this affect your measurement of V_{out}?

   The DMM is placed in parallel with the resistor being measured. If the DMM's resistance were only $25\,\text{k}\Omega$, it would draw a significant amount of current, acting as an additional parallel resistor that lowers the effective resistance of that branch. This would cause a smaller potential drop across the branch (since $\Delta V = I R_{\text{eff}}$ and R_{eff} is reduced), resulting in a measured V_{out} that is significantly lower than the true value.